FACULTY
OF MATHEMATICS
AND PHYSICS
Charles University

MASTER THESIS

Natalia Glazyrina

Identification of typical features of machine translation

Institute of Formal and Applied Linguistics

Supervisor of the master thesis: Jindřich Libovický Mgr. Ph.D.
Study programme: Computer Science
Study branch: Language Technologies and Computational Linguistics

Prague 2025

I declare that I carried out this master thesis on my own, and only with the cited sources, literature and other professional sources. Additionally, during the preparation of this thesis, I used ChatGPT-4o to improve my writing style. I declare that I used this tool in accordance with the principles of academic integrity. I checked the content and I take full responsibility for it. I understand that my work relates to the rights and obligations under the Act No. 121/2000 Sb., the Copyright Act, as amended, in particular the fact that the Charles University has the right to conclude a license agreement on the use of this work as a school work pursuant to Section 60 subsection 1 of the Copyright Act.

In .... date ....

Author's signature

This master's degree was a long and eventful journey, and it would not have been possible without people who supported me. First, I want to express my gratitude to my supervisor, Mgr. Jindřich Libovický, Ph.D., for being available and responsive, and for his patient guidance throughout the entire process.

I am deeply thankful my closest friend Artem — without his encouragement, I might never have even started this path.

My heartfelt thanks go to my beloved Danya, his wife Anya, and their daughter Alisa, whose endless optimism helped me through the most challenging moments of my studies.

I would like to thank my friend and fellow student Kirill, who constantly reminds me that life is full of possibilities.

I'm also grateful to my manager Jiří, who created a comfortable and supportive work environment that made it much easier to balance thesis writing with professional responsibilities.

And finally, my deepest gratitude goes to my mother Marina, whose unconditional support has always been my greatest source of strength and self-belief.

Title: Identification of typical features of machine translation

Author: Natalia Glazyrina

Institute: Institute of Formal and Applied Linguistics

Supervisor: Jindřich Libovický Mgr. Ph.D., Institute of Formal and Applied Linguistics

Abstract: Modern machine translation (MT) systems have advanced to the point where their outputs are often indistinguishable from human translations. Despite this, it is possible to train classifiers that can differentiate between human and machine-generated translations. In this work I focus on developing a classifier based on the pre-trained multilingual language model XLM-R to distinguish between human and machine translations, benchmarking its performance against a baseline classifier. Training data for this work were sourced from the WMT conference datasets from 2020 to 2022. To interpret the trained classifier, I apply an axiomatic attribution method. I annotate words in test sentences with linguistic features – such as Universal Part of Speech (UPOS) tags, grammatical case, and syntactic dependency relations – and analyze correlations between word-level attribution scores and these linguistic features.

Keywords: machine translation, axiomatic attribution, XLM-R, classification

Název práce: Identifikace typických rysů strojového překladu

Autor: Natalia Glazyrina

Ústav: Ústav formální a aplikované lingvistiky

Vedoucí diplomové práce: Jindřich Libovický Mgr. Ph.D., Ústav formální a aplikované lingvistiky

Abstrakt: Moderní systémy strojového překladu (MT) dosáhly úrovně, kdy jejich výstupy jsou často nerozlišitelné od lidských překladů. Přesto je možné natrénovat klasifikátory, které dokážou odlišit lidské překlady od strojových. V této práci se zaměřuji na vývoj klasifikátoru založeného na předtrénovaném vícejazyčném jazykovém modelu XLM-R, jehož úkolem je rozlišit mezi lidskými a strojovými překlady. Výkonnost tohoto modelu porovnávám s baseline klasifikátorem. Trénovací data použitá v této práci pocházejí z datových sad konference WMT z let 2020 až 2022. Pro interpretaci natrénovaného klasifikátoru využívám axiomatickou atribuční metodu. Slova ve větách testovacího souboru anotuji lingvistickými rysy — například univerzálními slovními druhy (UPOS), gramatickými pády a syntaktickými závislostními vztahy — a analyzuji korelace mezi atribučními skóre na úrovni slov a těmito lingvistickými rysy.

Klíčová slova: strojový překlad, axiomatická atribuce, XLM-R, klasifikace

Contents

Introduction

Recent advances in the Machine Translation (MT) field have led to the development of MT systems that can generate translations nearly indistinguishable from those produced by humans, when evaluated without a broader context. State-of-the-art neural MT models now achieve fluency and accuracy at a level that challenges human evaluators to differentiate between human-produced and machine-generated translations. However, while human judgment may be challenged by high-quality MT output, automated classifiers can be trained to distinguish between human and machine translations with high accuracy.

Previous research Ippolito et al. (2020) in this area has demonstrated promising results in distinguishing between human-authored text and language model-generated text using classifiers based on BERT Devlin et al. (2019). However, this research has focused primarily on English text, leaving space for further exploration into multilingual contexts and language pairs where translation quality and stylistic features vary significantly.

This thesis aims to extend the current research by developing a classifier to differentiate between human and machine translations specifically for English to Russian and English to Czech language pairs. This classification model was built upon a multilingual language model XLM-RoBERTa Conneau et al. (2020). Additionally, to gain insight into the model's decision-making process, I employ the integrated gradients methodSundararajan et al. (2017) – an interpretability technique that provides attributions for model predictions, thereby enabling analysis of the linguistic features that influence classification outcomes. The structure of this thesis is as follows:

- Chapter 1 provides a comprehensive overview of multilingual language models, including their architectures and fine-tuning approaches relevant to text classification task. This chapter also reviews machine translation evaluation methods and the axiomatic attribution technique.

- Chapter 2 reviews prior work Ippolito et al. (2020), which explores the classification of automatically generated versus human-authored text and sets a foundation for this study.

- Chapter 3 details the data collection and preparation process, as well as the model fine-tuning strategies used to train the text classifier for English-to-Russian and English-to-Czech translation pairs.

- Chapter 4 describes the interpretability analysis, applying integrated gradients to analyze attribution scores and exploring their correlation with linguistic annotations in the translated sentences.

1 Theoretical background

1.1 MT evaluation approaches

MT evaluation is the process of assessing the quality of machine-generated translations to determine how accurately and fluently they match human standards. Metrics of MT quality measure performance, which supports continuous improvements and helps developers target specific system weaknesses. MT evaluation methods fall into two primary categories:

1. Human Evaluation: Involves human judges assessing translations for adequacy, fluency, and meaning. For example, there is Direct Assessment (DA) Graham et al. (2015), where human evaluators rate translations directly on a scale (often from 0 to 100), providing a numeric score that reflects accuracy and fluency.

2. Automatic Evaluation: Uses algorithms to compare machine translations against reference human translations. Popular metrics include BLEU Papineni et al. (2002) and neural framework COMET Rei et al. (2020). In this work, I used COMET evaluation for dataset preparation step. A relatively recent metric, COMET (Cross-lingual Optimized Metric for Evaluation of Translation), uses a neural model to assess translations based on semantic similarity between machine output and human reference translations. COMET aims to capture nuanced meaning and contextual accuracy beyond what simpler metrics like BLEU can achieve. COMET is trained to predict various human judgements.

1.2 Multilingual language model XLM-RoBERTa

In this work, I fine-tune XLM-RoBERTa (XLM-R) as a binary classifier to distinguish between human and machine translations in Czech and Russian. I also rely on COMET metric for evaluation, which is based on the same underlying model. In this section I provide a brief overview of the XLM-R architecture and its key characteristics.

Multilingual language models (MLMs) such as mBERT Devlin et al. (2019) and XLM Lample and Conneau (2019) have advanced the field of cross-lingual understanding by pretraining large Transformer models across numerous languages. These models have demonstrated strong capabilities in cross-lingual transfer, achieving high performance across various benchmarks. Unsupervised representation learning has continually improved the benchmarks in natural language understanding, evolving from pretrained word embeddings to contextualized representations and Transformer-based language models. The current research in cross-lingual understanding has expanded these systems to support a broader range of languages and to facilitate the cross-lingual scenario, where a model trained in one language is effectively applied to others.

XLM-RoBERTa (XLM-R) Conneau et al. (2020) is a multilingual masked language model based on the Transformer architecture and pretrained on text data from 100 languages. XLM-R achieves state-of-the-art results on cross-lingual tasks, including classification, sequence labeling, and question answering. It performs competitively with monolingual models on benchmarks like GLUE Wang et al. (2019) and XNLI Conneau et al. (2018) and outperforms mBERT on cross-lingual classification tasks by a large margin in low-resource languages.

XLM-R is trained using a multilingual Masked Language Model (MLM) objective using monolingual data. It utilizes subword tokenization through SentencePiece, utilizing a unigram language model on raw text data. Training was conducted on the CommonCrawl Corpus, covering 100 languages. The model includes a large vocabulary of 250,000 tokens and comes in two configurations: XLM-R Base, with 12 layers, 768 hidden dimensions, and 270 million parameters; and XLM-R Large, with 24 layers, 1024 hidden dimensions, and 550 million parameters.

1.3 Classification evaluation metric

In the later chapters of this thesis, I perform binary classification to distinguish between human and machine-generated translations. To evaluate the performance of the classifier, I use the Receiver Operating Characteristic (ROC) curve and its associated Area Under the Curve (AUC) score as the primary evaluation metric.

The Receiver Operating Characteristic (ROC) curve is a graphical representation of a binary classifier's performance across varying decision thresholds. It plots the True Positive Rate (TPR), also known as sensitivity or recall, against the False Positive Rate (FPR), also called the fallout or type I error rate. These rates are derived from the confusion matrix, where:

- $TPR = \frac{TP}{TP+FN}$, representing the proportion of actual positives correctly identified.
- $FPR = \frac{FP}{FP+TN}$, representing the proportion of negatives incorrectly classified as positives.

The ROC curve is generated by adjusting the classification threshold (τ) and calculating the corresponding TPR and FPR at each step.

The quality of a ROC curve is often summarized using the Area Under the Curve (AUC). The AUC score ranges from 0.5 (random guessing) to 1.0 (perfect classification), where a higher AUC indicates better discriminatory ability.

One of the key advantages of the ROC curve is its independence from class imbalance. Since TPR and FPR are normalized rates—computed as proportions within the positive and negative classes, respectively—they are unaffected by the relative sizes of these classes. This was the reason for selecting this metric over accuracy in this work, given the relatively imbalanced nature of the dataset. Murphy (2022)

1.4 Interpretation of DNN

Integrated Gradient, introduced in Sundararajan et al. (2017), is a method for attributing the output of a deep neural network to its input features. Attribution helps interpret the decision-making process of machine learning models by identifying the contribution of individual input features to the final prediction. Attribution methods provide insight into how neural networks process data, helping debug models and interpret predictions.

Formal Definition of Attribution

Given a function $F : R^n \rightarrow [0, 1]$ and an input $x = (x_1, .., x_n) \in R^n$, the attribution $A_F(x, x^{`})$ at an input x relative to a baseline $x^{`}$ is defined as a vector $(a_1, .., a_n)$, where a_i quantifies the contribution of feature x_i to the model's prediction. The baseline $x^{`}$ represents a reference point, typically corresponding to an absence of features. This provides a reference point for measuring the impact of features in x.

Axiomatic Approach to Attribution

The authors identify two key axioms that any robust attribution method must satisfy:

1. Sensitivity: If an input differs from its baseline in a single feature, and this feature changes the output, the method must assign a non-zero attribution to that feature.

2. Implementation Invariance: Attributions should remain identical for functionally equivalent models, even if their internal implementations differ.

Most existing attribution methods fail to meet these axioms. For instance:

- Gradient-based methods: These methods compute attributions as the product of input features and the gradient of the model's output with respect to those features. However, they violate Sensitivity because the gradient can be zero in regions where the function flattens, even when the feature contributes to the output.

- Backpropagation-based methods: Techniques like Layer-wise Relevance Propagation (LRP) Binder et al. (2016) and DeepLift Shrikumar et al. (2017) compute attributions by backpropagating discrete gradients through the network. While they address some gradient issues, they violate Implementation Invariance because the chain rule doesn't work for discrete gradients and method can be sensitive to the network architectural changes.

Integrated Gradients Method

Integrated Gradients (IG) method is proposed as a solution that satisfies both Sensitivity and Implementation Invariance. It defines attributions as the integral of gradients along a straight-line path from the baseline $x^{`}$ to the input x. Mathematically, the attribution for the i-th feature is expressed as:

$$\text{IntegratedGrads}_i(x) = (x_i - x_i^{`}) * \int_0^1 \frac{\partial F(x^{`} + \alpha * (x - x^{`}))}{\partial x_i} d\alpha$$

The integral is typically approximated using Riemann sums, requiring the computation of gradients at intermediate points. In practice, 20 to 300 gradient evaluations is usually sufficient.

Additional Theoretical Properties

In addition to Sensitivity and Implementation Invariance, Integrated Gradients satisfy:

- Completeness: The total attribution sums to the difference between $F(x)$ and $F(x^‘)$, ensuring that all contributions are accounted for.
- Symmetry Preservation: Symmetric features (e.g., equally important input variables) receive identical attributions, a property that aligns with intuitive fairness in feature contributions.

Furthermore, Integrated Gradients are part of a broader class of Path Methods, which generalize attributions to gradients computed along any path between $x^‘$ and x. Path Methods are the only techniques that satisfy Sensitivity, Completeness, and Implementation Invariance simultaneously. Among Path Methods, IG is unique in preserving symmetry due to its reliance on the simplest possible path, the straight line.

Baseline Selection

A critical step in applying Integrated Gradients is the choice of the baseline $x^‘$, which serves as a reference point for attribution. The baseline should ideally correspond to a state where the model output is neutral or near zero. For example, a black image in vision models or a zero embedding in text models represents the absence of features. The authors emphasize that the choice of baseline significantly affects the attributions, as it defines the starting point for measuring feature contributions. A poorly chosen baseline can lead to misleading attributions or artifacts.

1.5 Related work

A number of recent studies have explored the task of distinguishing between human-written and machine-generated text. While much of this work has focused on general-purpose text generation (e.g., open-ended generation by language models), my work addresses a specific subdomain: machine translation. Nonetheless, insights from these studies are relevant, particularly in terms of evaluation methodology and model design. One notable example is the work of Ippolito et al. (2020), which investigates both human and machine performance in identifying machine-generated content.

Ippolito et al. (2020) investigate the problem of distinguishing between human-written and machine-generated texts, focusing on the detection capabilities of humans and automated systems. The study examines how different text generation strategies influence detection performance, and highlights differences in how humans and machines approach this task.

Specifically, the main goals of the study were to:

1. Explore how decoding strategies, such as top-k sampling, nucleus sampling, and untruncated random sampling, affect the ability to discriminate between human and machine text.

2. Evaluate the performance of human raters and automated discriminators based on excerpt length and sampling methods.

Experimental Setup

To address these goals, the study employs text generated by the GPT-2 language model using three decoding strategies:

- Untruncated random sampling: Tokens are sampled directly from the model's output distribution.

- Top-k sampling: Limits the token selection to the top k (here, k=40) most probable words.

- Nucleus sampling: Selects tokens from a subset where cumulative probabilities do not exceed p=0.96.

Human-written texts were drawn from the same data distribution as GPT-2's training set to minimize stylistic differences. Balanced datasets were created by combining these with the machine-generated texts. Positive examples represent machine-generated text, while negative examples represent human-written text. Excerpts vary in length, from 2 to 192 tokens, with additional variations based on the inclusion or absence of priming text.

For automatic detection, the authors evaluated the performance of a fine-tuned BERT model Devlin et al. (2019) against simpler baselines, such as logistic regression on bag-of-words and histogram-of-likelihood ranks. Fine-tuned BERT_{large} (cased) classifiers were trained separately on datasets generated with different sampling strategies. Their transferability was tested by evaluating their performance on decoding strategies other than those they were trained on.

For human detection, human raters, including untrained workers and expert raters (e.g., university students), were asked to identify machine-generated text across varying excerpt lengths. Their accuracy and agreement rates were analyzed to compare human ability to detect generated text against automatic detectors.

Results

1. Automatic Detection:

    - Fine-tuned BERT significantly outperformed simpler baselines, achieving over 80% accuracy on long excerpts generated by top-k sampling.
    - Discriminators trained on top-k samples showed poor transferability to other sampling methods, while those trained on nucleus sampling were more generalizable.

- Shorter excerpts resulted in lower detection accuracy, highlighting the importance of sequence length in machine detection.
- Decoding strategies that optimize for human-like text (e.g., top-k sampling) introduce statistical artifacts, making them easier for machine to detect.

2. Human Detection:

   - Human raters performed worse than automated detectors, with expert raters achieving a median accuracy of 74% and maximum accuracy of 85% on the longest excerpts.
   - Humans struggled most with top-k samples, which were easier for machines to detect due to their statistical anomalies (e.g., excessive use of high-likelihood words).
   - Even with training and guidance, human performance remained inconsistent, showing significant variability in accuracy and agreement.

Although this study focuses on text generation in English, it provides a useful baseline for understanding the strengths and weaknesses of human and automated detectors. In contrast, my work focuses specifically on machine translation, and uses a multilingual setup involving Czech and Russian. Unlike Ippolito et al. (2020), I fine-tune a multilingual language model (XLM-R) and incorporate linguistic feature analysis to interpret classification behavior.

2 Experiments

2.1 Dataset

In order to train and evaluate a classifier for distinguishing between human and machine translations, I first needed to construct high-quality, labeled datasets. These datasets serve as the foundation for all experimental work presented in this thesis. They are used not only to fine-tune the classification model but also to support detailed analysis of its performance across different languages and translation quality levels. Because the task focuses specifically on Czech and Russian, I constructed datasets that represent a diverse and realistic sample of translations in these two languages.

To develop a dataset for model training and evaluation, I selected data from the WMT conference held in 2020, 2021, and 2022, focusing on translations in English-Russian and English-Czech language pairs. The datasets include only the translated sentences without corresponding source English sentences, and translations from both languages are combined within each dataset. To ensure quality differentiation within the data, I applied two different quality assessment methods: Direct Assessment (DA), which is a human evaluation metric, and COMET, each serving distinct purposes in analyzing translation quality.

Dataset Preparation Using Direct Assessment (DA) Scores

For the DA-based approach, I used DA scores as a quality filter to divide machine translations into four distinct quality groups: (60–70), (70–80), (80–90), and (90–100). Additionally, I created a separate group for human translations, setting a quality score range of (80–100). This grouping strategy was driven by the need to avoid highly imbalanced datasets. Rather than creating corresponding quality groups for both machine and human translations (e.g., (70–80) for both MT and human translations), which could lead to significant imbalance due to differences in the number of examples available, I opted for separate quality ranges for human translations. Organizing the data into these five quality-based groups also allowed for a more granular analysis of the classifier's performance across varying levels of translation quality. Furthermore, identifying a specific quality range where the classifier performs reliably was essential for effectively conducting the interpretability analysis, as it ensured meaningful insights could be derived from the classifier's decisions. Table 2.1 provides a detailed overview of these DA datasets, including the proportions of each language and class within the dataset.

Dataset Preparation Using COMET Evaluation

In addition to DA scores, I used the COMET metric, specifically the Unbabel/wmt22-comet-da model Rei et al. (2022), to perform an alternative quality evaluation. COMET considers three sources of information—the original source sentence, a human reference translation, and the machine-translated candidate. For this part of the data preparation, I created three quality-based groups for machine

	Proportions of			
MT score in group	machine trans.	human trans.	ru sentences	cs sentences
60–70	65	35	43	57
70–80	70	30	42	58
80–90	75	25	42	58
90–100	80	20	41	59

Table 2.1 DA-based datasets characteristics.

translations, with ranges of (75–85), (85–95), and (95–100). This grouping was adapted based on the distribution of available sentences across these score ranges, leading to different thresholds from the DA-based groups. All available human translations were included without modification. Table 2.2 provides a breakdown of the COMET datasets, showing language and class proportions for each quality category.

	Proportions of			
MT score in group	machine trans.	human trans.	ru sentences	cs sentences
75–85	60	40	57	43
85–95	80	20	56	44
95–100	60	40	46	54

Table 2.2 COMET-based datasets characteristics.

This dual approach—using both DA and COMET for dataset preparation—enabled experimentation with datasets evaluated by human (DA) and automatic (COMET) metrics, enriching the training data and supporting a more nuanced analysis of classifier performance across different quality assessments. Additionally, all datasets, both DA- and COMET-based, were divided into training, testing, and validation subsets in an 80:10:10 ratio, ensuring a balanced and consistent structure for model training and evaluation. All datasets are available in the associated GitHub repository[1], as well as in electronic attachment A.4.

2.2 Baseline model

To build a baseline binary classifier for distinguishing between human and machine translations, I utilized the logistic regression model with a TF-IDF (Term Frequency-Inverse Document Frequency) representation.

[1]https://github.com/lack-of-purpose/thesis/tree/main/data

Logistic regression for binary classification

Logistic regression is a statistical method used for binary classification, where the goal is to predict the probability that a given input belongs to one of two classes. It models the relationship between a dependent binary variable and one or more independent variables using a sigmoid function, defined as:

$$\sigma(z) = \frac{1}{1 + \exp(-z)}$$

where z is a linear combination of the input features and model weights. The sigmoid function maps any real-valued input into a range between 0 and 1, which can then be interpreted as the probability of belonging to a specific class.

TF-IDF Representation

To represent the textual data, I employed TF-IDF Jones (1972), a feature extraction technique that transforms documents into high-dimensional, sparse vectors. Each vector dimension corresponds to a unique word in the vocabulary. The TF-IDF score of a word is computed as the product of two measures:

- Term frequency: Indicates how frequently a word appears in a document.
- Inverse document frequency: Reflects the rarity of the word across the entire dataset, reducing the importance of commonly used terms.

Implementation details

For the logistic regression model, I used the scikit-learn implementation, setting the penalty to L2 regularization (default parameter) to prevent overfitting, and the solver to lbfgs Jones (1972), which is well-suited for small to medium-sized datasets. The maximum number of iterations was adjusted to 300 to ensure convergence during training. The TF-IDF vectors were generated using scikit-learn's TfidfVectorizer function with its default parameters.

Evaluation metric

Given the imbalance in the dataset, the ROC-AUC (Receiver Operating Characteristic - Area Under the Curve) metric was selected for evaluation. Metrics like accuracy can be misleading because the model might predict the majority class most of the time and still achieve a high accuracy.

The results obtained are described in Tables 2.3 and 2.4.

2.3 Cross-lingual model fine-tuning

In Chapter 1, I outlined the reason for selecting XLM-R as the cross-lingual model for training the classifier, and here, I will describe the details of that process. Specifically, I opted for the XLM-R$_{base}$ version available in the Hugging Face model repository. This version was chosen due to its balance between computational

MT score in group	ROC-AUC score
60–70	0.516
70–80	0.537
80–90	0.521
90–100	0.512

Table 2.3 Baseline ROC-AUC classification score for DA-based datasets.

MT score in group	ROC-AUC score
75–85	0.643
85–95	0.520
95–100	0.544

Table 2.4 Baseline ROC-AUC classification score for COMET-based datasets.

efficiency and strong performance in cross-lingual understanding tasks, making it well-suited for fine-tuning on diverse datasets. To carry out the fine-tuning, I utilized the Hugging Face Transformers library, adapting the classification head of the model to the task at hand with datasets described in previous section.

For the task of hyperparameter optimization, I chose Optuna, a widely used Python package that supports automated hyperparameter tuning through various optimization algorithms. Optuna's Tree-structured Parzen Estimator (TPE) Bergstra et al. (2013) sampler was selected due to its effectiveness in navigating complex, high-dimensional parameter spaces. The TPE algorithm is a hyperparameter optimization technique designed to efficiently explore the search space by constructing and updating two Gaussian Mixture Models, denoted as l(x) and g(x). These models represent the probability distributions of promising and less-promising hyperparameter values, respectively. By maximizing the ratio between these models ($\frac{l(x)}{g(x)}$), TPE identifies hyperparameter configurations that are more likely to yield optimal performance.

I experimented with several hyperparameter configurations, particularly focusing on the parameters listed in Table 2.5: learning rate, warmup steps, and gradient accumulation steps. The source code for the hyperparameter tuning process is available in the GitHub repository accompanying this thesis, as well as in the electronic attachment A.4.

Hyperparameter	Value
Learning rate	[1e-07, 1e-06]
Learning rate scheduler	Linear
Weight decay	0
Warmup steps	[500, 700, 1000]
Gradient accumulation steps	[4,8]
Batch size	8
Number of epochs	15
Optimizer	AdamW

Table 2.5 Hyperparameters in experiments.

To assess the training performance, I analyzed the evaluation loss and ROC-

AUC metrics across various training runs. Figures 2.1 and 2.2 show these metrics for the DA dataset (90-100), where the classifier achieved its highest performance. Further visualizations for other DA dataset experiments are included in Attachment A.1.

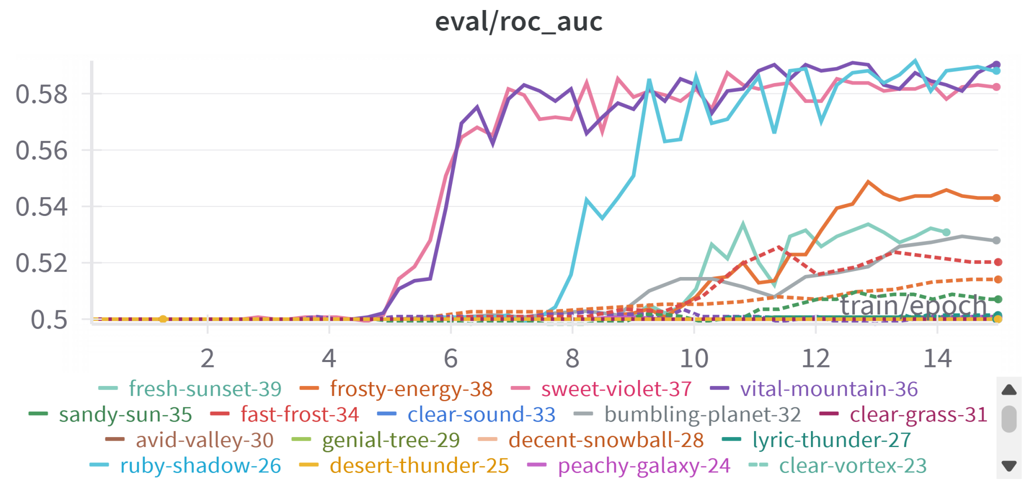

Figure 2.1 ROC-AUC score for (90–100) DA dataset.

Figure 2.2 Evaluation loss for (90–100) DA dataset.

Initially, I hypothesized that lower machine translation scores would correlate with improved classifier performance, anticipating that less accurate MT examples might better highlight the distinctions between human and machine-generated translations. However, as shown in Table 2.6, this hypothesis was not confirmed by training outcomes on the DA datasets.

MT score in group	ROC-AUC score
60–70	0.54
70–80	0.58
80–90	0.55
90–100	0.58

Table 2.6 Maximum achieved ROC-AUC score for DA datasets.

Based on that outcome, I shifted to training with datasets evaluated using the COMET metric. In this case, the best classifier performance was achieved when fine-tuning on the lower-score COMET dataset (75-85). The resulting model can be found in the associated GitHub repository[2]. Figures 2.3 and 2.4 illustrate the evaluation loss and ROC-AUC metrics for this dataset. Further visualizations for other COMET dataset experiments are included in the Attachment A.1.

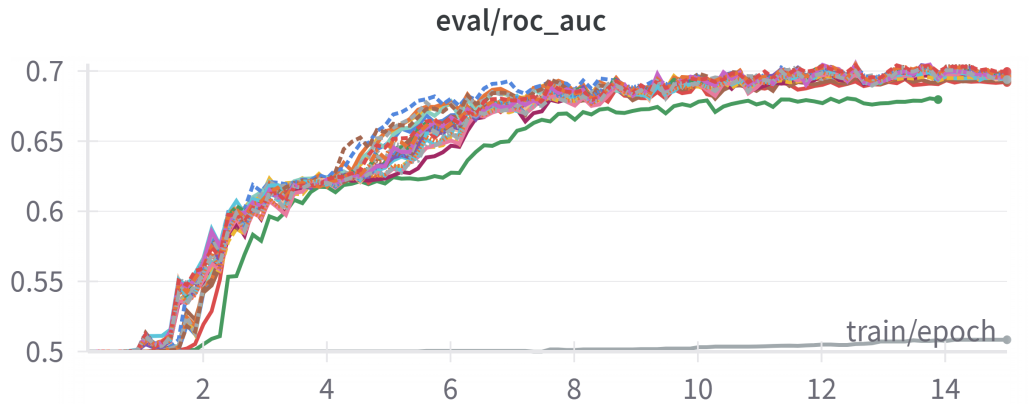

Figure 2.3 ROC-AUC score for (75–85) COMET dataset.

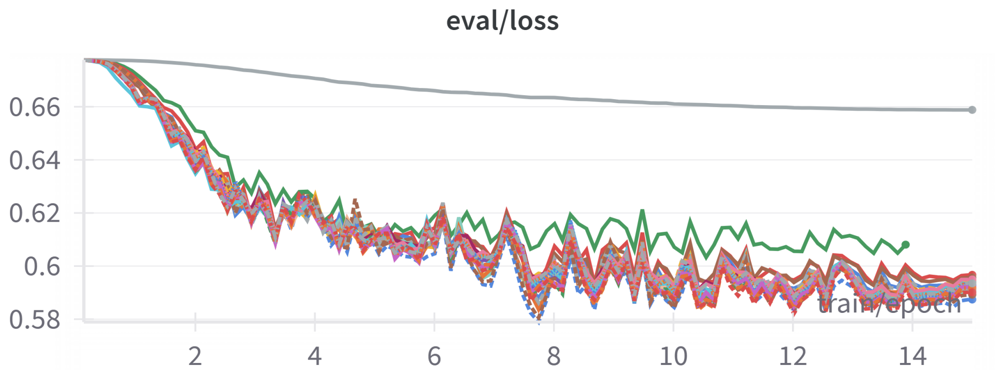

Figure 2.4 Evaluation loss for (75–85) COMET dataset.

Table 2.7 substantiates the initial hypothesis: fine-tuning on COMET-based datasets demonstrates the improvement in the classifier's ability to differentiate between human and machine translation as MT quality decreases, supporting the proposed approach for dataset selection based on quality metrics.

MT score in group	ROC-AUC score
75–85	0.705
85–95	0.583
95–100	0.572

Table 2.7 Maximum achieved ROC-AUC score for COMET datasets.

The best model was trained using the most promising set of hyperparameters listed in Table 2.8.

[2]https://github.com/lack-of-purpose/thesis/tree/main/xlm-roberta-base-comet-75-85

Hyperparameter	Value
Learning rate	0.000000985796741515975
Learning rate scheduler	Linear
Weight decay	0
Warmup steps	700
Gradient accumulation steps	4
Batch size	8
Number of epochs	15
Optimizer	AdamW

Table 2.8 Best model hyperparameters.

The resulting model was trained on Tesla P100-PCIE-16GB GPU and achieved a ROC-AUC score of 0.72 on the validation set and 0.70 on the test set. Figures 2.5 and 2.6 illustrate the evaluation loss and ROC-AUC score during training.

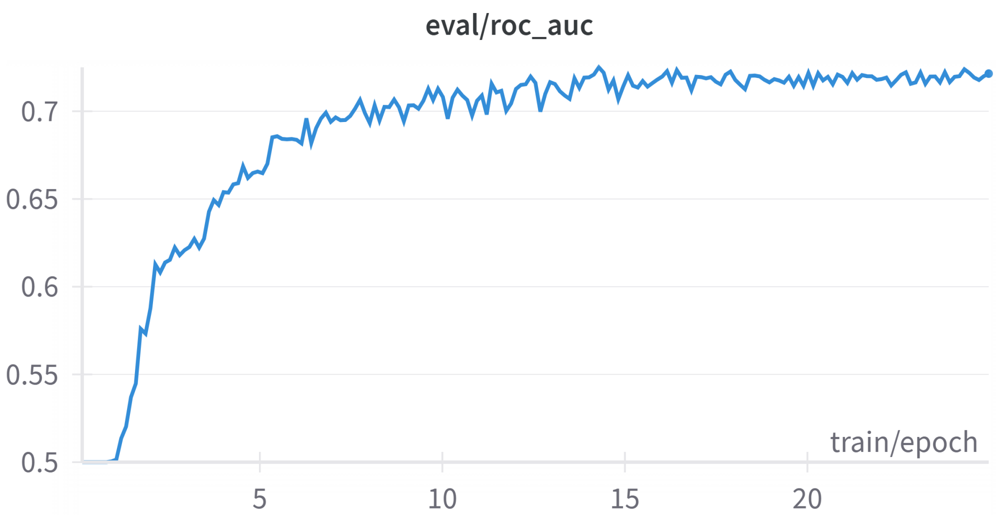

Figure 2.5 ROC-AUC score for the best model.

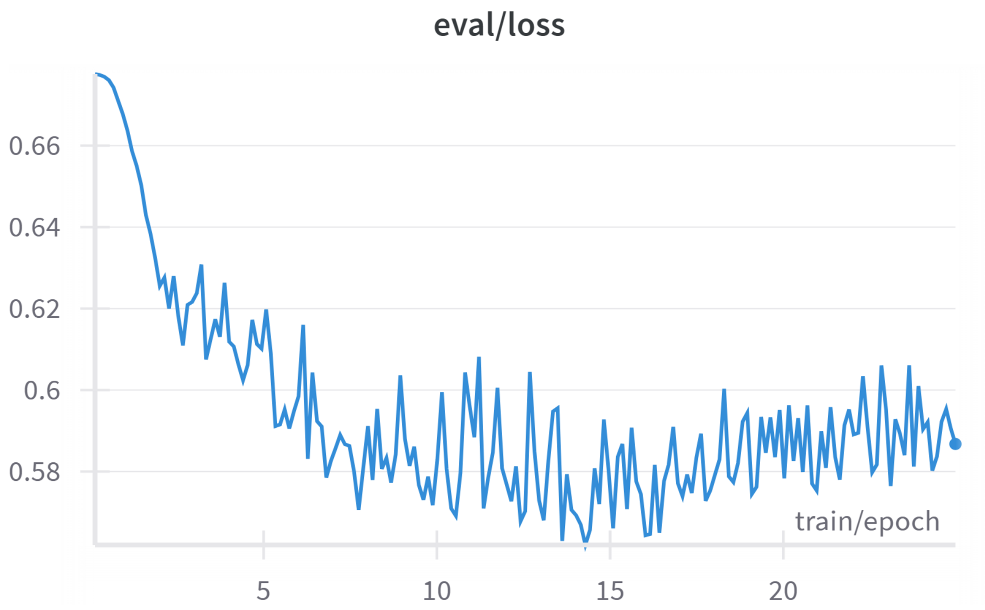

Figure 2.6 Evaluation loss for the best model.

3 Analysis

This chapter presents a comprehensive analysis of the fine-tuned XLM-R model and the linguistic factors potentially influencing its predictions. The goal is to gain a deeper understanding of the model's decision-making process and to identify patterns that may explain correct and incorrect classifications in Czech and Russian languages. First, the Model interpretation section describes how the Integrated Gradients method Sundararajan et al. (2017) was applied to visualize and quantify subword-level attribution scores. Next, the Automatic linguistic annotation section outlines the tools and methods used to extract linguistic features from the dataset. These features form the basis for the Statistical analysis, where approaches such as ordinary least squares (OLS) regression and point-biserial correlation were used to investigate relationships between linguistic phenomena and attribution scores. Finally, the Findings section summarizes key insights obtained from the analysis, highlighting differences between correctly and incorrectly classified examples in Czech and Russian.

3.1 Model interpretation

To interpret the classification decisions of the best-performing model, I utilized an axiomatic attribution approach implemented in the transformer-interpret Python package[1]. This package leverages the PyTorch Captum library[2], specifically its implementation of the Integrated Gradients method Sundararajan et al. (2017). Additionally, the package provides a convenient wrapper to facilitate its application with Hugging Face transformer models[3].

While the transformer-interpret package works well with many transformer-based models, it encountered compatibility issues when applied to the XLM-R model, producing the following error: "IndexError: index out of range in self". To be able to use it I had to make changes in the package. Further details provided in Attachment A.2

The transformer-interpret package call for a specific sentence returns subword-level attribution scores. An example of these subword-level scores is presented in Table 3.1.

[1]https://github.com/cdpierse/transformers-interpret
[2]https://captum.ai/
[3]https://huggingface.co/docs/transformers/index

Subword	Score
_Problém	-0.23
_s	0.007
_teplot	-0.053
ou	0.71
_vody	0.24
_v	0.26
_boj	0.0006
ler	-0.072
u	-0.065
_a	-0.25
_van	-0.087
ou	0.35
_	0.16
.	0.30

Table 3.1 Subword-level attribution scores.

For subsequent analysis, I aggregated the subword scores into word-level scores. Instead of using a simple average, which could underestimate a word's importance, I decided to assign each word the maximum score among its constituent subwords. This approach ensures that a word's contribution is not diminished, especially in cases where certain subwords play a disproportionately significant role in the model's decision.

Additionally, I created visualizations of subword attributions. These visualizations, saved as HTML files, allow for an interactive exploration of how individual subwords influence the model's predictions. An example of this visualization is illustrated in Figure 3.1.

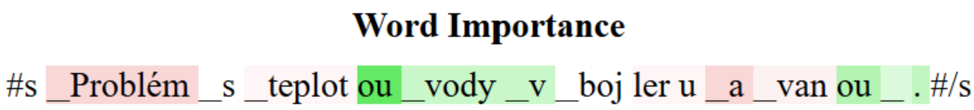

Figure 3.1 Visualization of subword-level attribution.

3.2 Automatic linguistic annotation

In contrast to Ippolito et al. (2020), where the characteristics of machine-generated and human-authored texts were analyzed from a statistical perspective, I chose to explore translations based on words' linguistic features. To better understand which of them might influence the model's predictions and attribution scores, I used different automatic approaches to extract information such as part-of-speech tags, dependency relations, morphological categories, and named entities. It served as the basis for the statistical analysis, where I explored potential correlations between linguistic characteristics and the model's output.

I conducted linguistic annotations using the UDPipe tool to extract Part-of-Speech (POS), dependency relations, case, number, gender and animacy.

UDPipe is a state-of-the-art language processing tool based on Universal Dependencies (UD), a framework for consistent grammatical annotation across languages. It provides a robust pipeline for linguistic analysis, including tokenization, POS tagging, lemmatization and dependency parsing Straka (2018). UDPipe supports a wide variety of languages which makes it well-suited for cross-lingual study.

UDPipe operates using trained models for various UD treebanks. These models are trained on annotated datasets and capture the grammatical nuances of each language. When analyzing a text, UDPipe processes it sequentially through its pipeline:

1. Input: The raw text is provided as input.
2. Tokenization and Sentence Splitting: The text is segmented into tokens and sentences.
3. Annotation: The tool assigns POS tags, lemmas, and syntactic dependencies to the tokens.
4. Output: The annotated text, including linguistic features and dependency trees, is returned in a CoNLL-U format[4].

For the analysis, I used the online UDPipe service, employing the pre-trained models czech-pdt-ud-2.15-241121 and russian-syntagrus-ud-2.15-241121 to annotate texts in Czech and Russian, respectively.

For further analysis, I also extracted Named Entities (NE) using different tools for Czech and Russian sentences: NameTag 3[5] and Natasha[6] python package, respectively.

NameTag 3 Straková et al. (2019) is an advanced open-source tool designed for recognizing both flat and nested named entities in text. It detects proper names and categorizes them into predefined groups, such as persons, locations, organizations, and more. This state-of-the-art tool supports 17 languages, including Czech, English, German, and Ukrainian. It is based on a fine-tuned large language model (LLM) that uses either a softmax classification head for identifying flat NER or a sequence-to-sequence decoding head for identifying nested NER.

Natasha is a comprehensive open-source Python library specifically designed for natural language processing (NLP) tasks in the Russian language. It offers a suite of tools that address various linguistic challenges, including tokenization, sentence segmentation, morphological tagging, lemmatization, syntax parsing, and named entity recognition (NER). In the realm of NER, Natasha provides the functionality to identify and classify proper names within Russian text into predefined categories such as persons (PER), locations (LOC), and organizations (ORG). The library integrates deep learning models, particularly through its Slovnet component, to achieve high-quality recognition of these entities.

Annotated datasets are available in the electronic attachment A.4.

[4]https://universaldependencies.org/format.html
[5]https://lindat.mff.cuni.cz/services/nametag/info.php
[6]https://github.com/natasha/natasha

3.3 Statistical analysis approaches

In this work, two complementary statistical methods were used to analyze the relationship between linguistic features and word-level attribution scores: Ordinary Least Squares (OLS) regression with categorical predictors and point-biserial correlation. These methods were chosen to provide different yet reinforcing perspectives on how categorical linguistic features relate to continuous attribution scores.

Ordinary Least Squares (OLS) Regression with Categorical Predictors

The primary goal of using OLS regression Nguyen et al. (2023) was to determine how different categories of a linguistic feature, such as Universal Part-of-Speech (UPOS) tags, relate to the continuous attribution score. OLS regression provides an estimate of how each level of a categorical variable, such as NOUN, VERB, or ADJ, differs from a reference level in predicting a numeric outcome.

Rather than using a single reference category, I applied sum contrast coding Vasishth et al. (2022), which allows for a comparison of each category's effect relative to the grand mean. This approach ensures that the estimated coefficients represent deviations from the overall mean rather than a specific reference category, thus offering a more holistic interpretation of the effect of each category.

The model was constructed using the formula score ˜ C(upos, Sum), where score represents the dependent variable (word-level attribution score) and C(upos, Sum) represents the categorical predictor (UPOS tag) with sum coding. Once the model was fitted, its summary was examined to interpret the significance and direction of each coefficient.

To determine the significance of each UPOS category, I relied on a p-value threshold of 0.05, retaining only those categories that met this criterion. The sign of the estimated coefficient was then examined to infer whether the category had a positive or negative association with the attribution score.

The key advantage of this approach is that it provides direct interpretability, as the coefficients quantify how each category differs from the grand mean.

The Python code used for this approach is included in the electronic attachment A.4 for reference.

Point-Biserial Correlation

To further examine the relationship between individual linguistic features and attribution scores, I used point-biserial correlation Tate (1954), a special case of Pearson correlation designed for situations where one variable is continuous and the other is binary. This method was particularly useful for assessing the strength and direction of relationships at the individual category level without the constraints of a reference coding scheme.

To apply this method, categorical features were first one-hot encoded, ensuring that each category was represented as a separate binary feature. For each category, a point-biserial correlation coefficient was computed, measuring the strength of association between the presence of the category and the attribution score. The statistical significance of the correlation was determined using a p-value threshold of 0.05, and the direction of association was assessed based on the sign of the correlation coefficient.

Point-biserial correlation is particularly informative because it provides a straightforward measure of association between a categorical feature and the outcome variable. Unlike OLS regression, it does not rely on contrast coding or multiple comparisons, making it a useful complementary approach for validating findings.

By utilizing both OLS regression and point-biserial correlation, I ensured a robust, cross-validated approach to statistical analysis. These methods offer complementary insights: OLS regression with sum coding treats categorical features as multi-level factors, estimating the unique contribution of each level relative to the grand mean, while point-biserial correlation examines each feature independently, offering a simple binary comparison. The consistency of results across methods strengthens the reliability of the findings, as significant features identified by both methods can be interpreted with greater confidence.

However, in some cases point-biserial correlation approach revealed more statistically significant features. It can be explained by the fact that point-biserial correlation provides an isolated comparison. It is computed separately for each category (e.g., "Is UPOS = ADJ?") against the continuous outcome (score). This means each category's correlation is unaffected by the distribution or variance of other categories. In OLS, if two categories tend to co-occur in certain contexts (e.g., some morphological tags might frequently appear together), the model might split the explanatory power between them. This can reduce each category's individual significance.

3.4 Findings

For the analysis of linguistic features which characterize machine-translated and human-translated texts, I divided test part of dataset into two groups: correctly and incorrectly classified. In both those groups, I analyzed separately features for Czech and Russian texts.
All linguistic features mention in the following sections are listed in Attachment A.3, along with their abbreviations and explanations.

3.4.1 Correctly classified Czech texts

Machine-translated (MT) sentences

According to results, presented in Table 3.2, the choice of MT class was negatively impacted by `acl:relcl` (relative clause modifier) dependency relation, which means that in the presence of relative clause modifier the classifier is less likely to classify the sentence as machine-translated. In other words, the presence of relative clauses seems to signal a more human-like translation. Czech relative clauses can be syntactically complex and require nuanced word order and agreement rules. Machine translation (MT) systems often simplify or rephrase subordinate structures instead of preserving their complexity. Therefore, encountering a well-formed Czech relative clause may indicate a higher likelihood of human translation.

Linguistic feature	Machine-translated	Human-translated
UPOS	–	NUM (+), CCONJ, NOUN, PRON (-)
Dependency relation	acl:relcl(-)	nummod (+), advmod:emph, cc, ccomp, obj, obl (-)
Case	–	Acc (-)
NER	T|tm (-)	T|td (+)
Gender	(Masc,Neut) (-)	–
Number	Plural (-)	–

Table 3.2 Czech text's features in correctly classified examples.

Similarly, the presence of a word referring to months - NER=`T|tm` - (e.g., květnu, října, etc.) also decreases the probability of a machine-translated classification. Machine-translated sentences may handle temporal expressions in a more formulaic or standardized manner, whereas human translators tend to select contextually appropriate variations. For instance, human translators may choose different grammatical cases (e.g., v květnu vs. během května). This variety might stand out as human-like", reducing the probability of a machine-translation label.

The classifier's probability of assigning a sentence to the MT category was also negatively impacted by gender label `(Masc, Neut)`. In the test dataset, there are mainly pronouns and determiners having the gender label `(Masc,Neut)`, because they have the same form for masculine and neuter gender (e.g. toho, svém, vašeho, etc.). Correct application of Czech gender agreement, particularly for masculine and neuter forms, requires precise adjective-noun or pronoun-noun matching. When gender agreement is used correctly in context, it suggests a more nuanced application of Czech morphology, making the classifier less likely to assign the sentence to the machine-translated category.

Similarly, correct plural usage, including case and number agreement in plural forms, is another subtle feature impacting classification. Machine translation systems occasionally make errors in morphological choices for plural forms or fail to maintain consistency throughout a sentence. In contrast, fluent and consistent plural usage is a strong indicator of human translation.

Overall, Czech features a complex system of agreement rules, where nouns, adjectives, and pronouns must match in gender (masculine, feminine, neuter) and number (singular, plural). These grammatical categories are important markers of fluent translation and, when applied correctly, decrease the probability of a sentence being classified as machine-translated.

Human-translated (HT) sentences

The classification of a sentence as human-translated was positively impacted by the presence of UPOS = `NUM` (numerical expressions) and dependency relation=`nummod` (numeric modifiers), as presented in Table 3.2,. The presence of numerical tokens or numerical modifiers correlates with a higher likelihood of human translation. Human translators tend to match the style and formatting of numbers from the source text in a more natural manner, for instance, choosing between spelled-out and digit forms or ensuring correct grammatical agreement of numbers with nouns in Czech. Additionally, human translators may preserve numerical information in a more idiomatic way, adjusting case endings or modifying word order to ensure fluency.

The feature NER=`T|td` (Time:Days) also positively impacted the classifier's decision to assign a sentence to the HT category. Similar to months, day expressions (e.g., pondělí, úterý) are often handled in a more idiomatic and context-sensitive manner by human translators, who adjust prepositions, inflection, and word order accordingly.

In contrast, several linguistic features negatively impacted the classification of a sentence as human-translated:

- Part of speech (UPOS): `CCONJ` (coordinating conjunction), `NOUN`, `PRON` (pronoun)
- Dependency relations: `advmod:emph` (emphasizing word, intensifier), `cc` (coordinating conjunction), `ccomp` (clausal complement), `obj` (object), `obl` (oblique modifier)
- Case: `Acc` (Accusative)

A negative coefficient for these features suggests that their presence decreases the probability of a sentence being classified as human-translated.

Machine translation systems tend to insert additional conjunctions or directly link clauses using simpler structures (e.g., a, ale), rather than using a wider range of connectors that a human translator might choose.

Pronouns (`PRON`) and Accusative case (`Acc`) may also be used in a more formulaic or literal manner in machine translations. For example, an MT system may translate an English pronoun into a Czech pronoun consistently, even when natural Czech would omit or rephrase it. In the test dataset I use for analysis, many pronouns were reflexive (se/si), suggesting that reflexive pronouns were overrepresented in the machine-translated training data, leading the classifier to associate them with machine translation.

Dependency labels such as `ccomp` and `obl` (clausal complements, oblique modifiers) appear more frequently in direct or literal translations, as machine translation may systems systematically map source syntactic structures into Czech. In contrast, human translators often restructure complex complements or rephrase objects to enhance fluency, reducing the frequency or prominence of these labels in human translations.

Finally, `advmod:emph` (emphatic adverbs) appears more frequently in machine-translated outputs. For example, a system may not always capture Czech's subtle discourse particles correctly and instead produce an over-reliance on certain adverbs.

Linguistic feature	Machine-translated	Human-translated
UPOS	ADJ, NOUN (+), DET,PART (-)	–
Dependency relation	amod, appos, flat:foreign (+), conj, det, fixed, mark (-)	flat:name, nmod (+)
NER	ORG (+)	LOC (+)

Table 3.3 Russian text's features in correctly classified examples.

Key takeaways

The analysis highlights several key differences between machine-translated and human-translated sentences. Analyzed human translations are characterized by:

- A potentially richer or more context-sensitive use of numerical expressions
- More complex relative-clause constructions (as indicated by the negative impact of `acl:relcl` on machine classification)
- A more flexible representation of time expressions

Machine translations, on the other hand, tend to:

- Use certain syntactic structures (e.g., direct objects, `ccomp`, pronouns) in a more literal or higher-frequency manner than humans do
- Either avoid or handle complex structures (such as relative clauses) less often or in a more standardized form

3.4.2 Correctly classified Russian texts

Machine-translated (MT) sentences

The classification of a Russian sentence as machine-translated was positively influenced by the presence of `ADJ` (Adjective) and `NOUN` (Noun) parts of speech, as shown in Table 3.3 English-to-Russian machine translation systems often map English adjectives and nouns directly, frequently preserving word order and morphological patterns in a straightforward manner. A high frequency of adjective-noun pairs—particularly in default positions—or a prevalence of bare nouns may indicate a less context-sensitive, more literal machine translation approach.

This observation is reinforced by the presence of the `amod` (adjectival modifier) dependency relation, which reflects a high number of direct adjective-noun constructions.

Additionally, the flat:foreign dependency relation, which marks foreign words, was another indicator of machine translation. Machine translation systems often leave foreign words or transliterated terms unchanged, particularly when dealing with proper names or brand names. This results in repeated foreign fragments, which can be a feature of MT-generated text.

A similar effect was found with NER=ORG (Organization), where machine-translated output systematically preserves organization names (e.g., company names, institutions) in either direct transliterations or unchanged English forms. In contrast, human translators are more likely to adapt these names to Russian conventions, such as adding suffixes, abbreviations, or using widely recognized Russian equivalents. Thus, the direct presence of an ORG entity in an unaltered form can be a characteristic of machine translation.

Conversely, several linguistic features negatively impacted the classification of a sentence as machine-translated, including the UPOS: DET (Determiner) and PART (Particle) categories, as well as the conj (Conjunct), det (Determiner), fixed (Fixed Multi-Word Expression), and mark (Marker) dependency relations.

Russian language does not have definite or indefinite articles (the, a). If a word is tagged as a determiner (UPOS=DET, deprel=det), it likely represents demonstrative determiners (этот) or relative determiners (которая). Machine translation systems may omit or inconsistently apply such determiners.

Similarly, certain Russian particles (PART) are often used for emphasis, politeness, or idiomatic expression — features that are frequently misrepresented or entirely absent in machine-translated text. Human translators naturally incorporate these linguistic elements, making their presence a strong signal of human translation.

Machine translation systems also tend to handle conjunctions (conj) in a more rigid or oversimplified manner, whereas human translators use more flexible and idiomatic coordinating structures. Additionally, the fixed dependency label, which is used for multi-word expressions, was another negative marker for MT classification. Human translators are more likely to use and preserve fixed Russian expressions appropriately, whereas machine translation systems may produce literal, word-for-word renderings that lack natural fluency.

Human-translated (HT) sentences

The choice of human-translated class was positively impacted by the flat:name (names), nmod (nominal modifier), and NER = LOC (Location) dependency relations (see Table 3.3).

Human translators tend to render personal names and multi-element proper names more accurately, including correct case endings. The presence of the flat:name pattern is thus a strong indicator of human involvement, as it reflects greater attention to grammatical accuracy.

The impact of nmod dependency relation can be explained the following way. Humans might restructure noun phrases in a more contextually appropriate way, using genitive phrases or other nominal constructions that reflect Russian's rich case system.

The impact of the nmod (nominal modifier) dependency relation can be explained by the way human translators restructure noun phrases in a more contextually appropriate manner. Rather than directly mirroring English syntactic structures, they tend to use genitive phrases and other nominal constructions to produce more fluent and natural translations.

Another key feature distinguishing human translation was the handling of location names (NER= LOC). Location names can be translated in multiple ways, including transliteration, transliteration with added Russian suffixes, or full adaptation into Russian equivalents (e.g., Нью-Йорк vs. New York). Human translators frequently recognize and use conventional Russian equivalents for foreign locations (Пекин for Beijing, Милан for Milan), whereas machine translation systems are more likely to produce direct transliterations or leave location names unchanged. The correct adaptation of such entities can serve as a marker of human translation.

Key takeaways

The analysis highlights key differences between machine-translated and human-translated Russian sentences:

Machine-translated sentences tend to exhibit:

- A higher frequency of literal morphological and lexical mappings from English, particularly straightforward adjective-noun pairs
- The preservation of foreign words and organization names in their original or transliterated form (e.g., flat:foreign, NER= `ORG`)

Human-translated sentences, in contrast, demonstrate:

- More nuanced usage of Russian morphology, including greater reliance on nominal modifiers (`nmod`) and idiomatic noun-phrase restructuring
- More appropriate adaptation of proper names and locations, avoiding direct transliterations where possible (flat:name, NER= `LOC`)
- A richer use of particles, demonstratives, and fixed expressions, contributing to more natural Russian sentence structures

3.4.3 Incorrectly classified Czech texts

This section provides an interpretive analysis of instances where the classifier incorrectly assigned labels—either classifying human-translated sentences as machine-translated or vice versa. The analysis explores the specific linguistic features that influenced these incorrect classifications, either positively (pushing the classifier toward the incorrect label) or negatively (pushing it away from the incorrect label).

Linguistic feature	Machine-translated	Human-translated
UPOS	ADJ (+), ADP, CCONJ, DET (-)	–
Dependency relation	amod (+), advcl, advmod, case (-)	flat (+), obl:arg, advmod:emph, cop (-)
NER	–	P|ps, i_ (+), gc (-)
Gender	Masc (+)	(Masc,Neut) (-)
Animacy	–	Anim (+)
Number	–	Plural (-)

Table 3.4 Czech text's features in incorrectly classified examples.

Human-translated sentences, classified as Machine-translated

The incorrect classification of human-translated sentences as machine-translated was positively influenced by the presence of the following linguistic features (Table 3.4):

- UPOS: `ADJ` (Adjective)
- Dependency relation: `amod` (adjectival modifier)
- Masculine gender

The presence of adjectives (`ADJ`) and adjectival modifiers (`amod`) appears to have influenced the classifier toward assigning the "machine-translated" label. One possible explanation is that in genuine machine translations, adjective-noun structures are often preserved or simplified. For instance, machine translation systems frequently map English adjectives to Czech adjectives in a direct manner, rather than adapting them to Czech's more flexible syntactic structures. Consequently, the classifier may have "learned" that frequent `ADJ`–`amod` patterns are indicative of machine translation. This pattern aligns with a similar finding observed in Russian machine translations, where straightforward adjective-noun pairings also signaled MT output.

In contrast, the following linguistic features reduced the likelihood of the classifier assigning the "machine-translated" label:

- UPOS: `ADP` (Adposition), `CCONJ` (Coordinating Conjunction), `DET` (Determiner)
- Dependency relations: `advcl` (Adverbial Clause Modifier), `advmod` (Adverbial Modifier), `case` (Case Marking)

The presence of adpositions (`ADP`) and corresponding case markings (`case`) suggests a richer and more precise use of grammatical structures, which is more characteristic of human translations. Similarly, a nuanced use of adverbial clauses (`advcl`) and adverbial modifiers (`advmod`) may contribute to greater fluency, making these features more indicative of human translation.

Determiners (`DET`) in Czech, like in Russian, do not function as direct equivalents to English articles but instead include demonstrative determiners (e.g., tento) and relative determiners (e.g., kteří). Human translators tend to use these determiners more precisely for emphasis or clarity. This distinction likely led the classifier to associate their presence with human translations.

Machine-translated sentences, classified as Human-translated

The incorrect classification of machine-translated sentences as human-translated was positively influenced by the following features (Table 3.4):

- Dependency relation: `flat`
- NER: `P|ps` (Personal Name - Surname), `i_` (Institution)
- Animacy: `Animate`

The classifier likely identified well-structured named entities, such as institution names (`i_`) and surnames (`P|ps`, `flat`), as indicative of human translation. While human translators typically ensure proper handling of such names, modern machine translation systems can also handle those names properly. This may have led the classifier to erroneously associate such polished name-handling with human translation.

Similarly, the presence of animate morphological marking—such as noun/adjective endings or pronoun usage distinguishing animate from inanimate forms—pushed the classifier toward labeling the sentence as human-translated. Since correctly handling animacy in Czech requires syntactic and morphological awareness, the classifier may have assumed that its accurate application was a hallmark of human translation. However, contemporary machine translation systems are capable of preserving or deriving correct animacy from context, potentially misleading the classifier.

Conversely, the following features negatively impacted the classifier's likelihood of assigning the "human-translated" label:

- Dependency relations: `obl:arg` (Oblique Argument), `advmod:emph` (Emphasizing Word/Intensifier), `cop` (Copula)
- NER: `gc` (States)
- Gender: Masculine/Neuter `(Masc,Neut)`
- Plural number

Machine translation (MT) systems often translate argument structures in a direct, source-oriented manner, resulting in oblique arguments (`obl:arg`) that follow a systematic but sometimes rigid pattern. In contrast, human translators are more likely to rearrange or paraphrase these arguments to achieve a more natural and idiomatic flow in the target language. Machine translations, however, tend to preserve the original argument slots, leading to predictable and sometimes unnatural oblique structures.

Similarly, MT systems frequently handle state and country names in a uniform and inflexible way, often defaulting to the nominative form and neglecting appropriate case endings required by Czech grammar. This lack of morphological adaptation can contribute to awkward or incorrect phrasings that are less characteristic of human translation.

Issues with Czech morphology are also evident in the handling of plural forms and masculine/neuter gender forms. Machine translation systems may struggle with grammatical agreement, leading to inconsistencies in number and gender that a human translator would naturally resolve through contextual adaptation.

A particularly distinguishing feature of machine translation is the presence of `advmod:emph` (emphasizing adverbs and intensifiers), which was statistically significant in the analysis of correctly classified Czech texts. This feature negatively impacted the classification of human translations, reinforcing its association with machine-translated output. Machine translation systems may either overuse intensifiers, mapping them too directly from the source language, or fail to capture their nuanced placement, resulting in unnatural or awkward emphasis in the target text.

Key takeaways

The model struggled to decide if it is machine or human translation for the following sentences:

- Human translations, labeled as "machine": text contained straightforward adjective–noun constructions (`ADJ` + `amod`) and did not feature the variety of prepositions, conjunctions, determiners, or complex adverbial structures

- Machine translations, labeled as "human": text handled proper names and institutions in a seemingly polished or consistent way

3.4.4 Incorrectly classified Russian texts

This section provides an interpretive analysis of instances where the classifier incorrectly assigned labels to Russian translations—either classifying human-translated sentences as machine-translated or vice versa. The analysis examines the linguistic features that contributed to these misclassifications, either by positively influencing the classifier toward the incorrect label or by negatively impacting its likelihood of making the correct classification.

Linguistic feature	Machine-translated	Human-translated
UPOS	ADJ (+)	INTJ (+), CCONJ, DET, VERB (-)
Dependency relation	fixed (+), advmod, mark (-)	flat:name, discourse (+), acl:relcl, det, fixed (-)
NER	–	PER (+), NONE, ORG (-)
Animacy	–	Anim (+)
Number	Plural (+)	–

Table 3.5 Russian text's features in incorrectly classified examples.

Human-translated sentences, classified as Machine-translated

The incorrect classification of human-translated sentences as machine-translated was positively influenced by the following features (Table 3.5):

- UPOS: `ADJ` (Adjective)
- Dependency relation: `fixed` (fixed multi-word expression)
- Number: Plural

The presence of adjectives (`ADJ`) and fixed multi-word expressions (`fixed`) led the classifier to assign the "machine-translated" label. As noted in previous sections, machine translation systems often produce direct and literal adjective-noun mappings. Additionally, certain repeated or standardized phrases might get annotated as fixed, and the classifier may have learned that "unchanged multi-word chunks" are typical for machine translations. However, human translators also frequently use straightforward adjective constructions, particularly when preserving descriptive details from the source text.

Conversely, the classifier was less likely to assign the machine-translated label when the following features were present:

- Dependency relation: `advmod` (Adverbial Modifier), `mark`(Subordinating Marker)

The presence of adverbial modifiers (`advmod`) and subordinate markers (`mark`) reduced the classifier's likelihood of labeling the sentence as machine-translated. One possible explanation is that the classifier associates a more flexible use of adverbs and subordinate clauses — both of which enhance sentence complexity and nuance — with human translation. Sentences with fewer adverbs or subordinate conjunctions may appear more literal or simplified, leading the classifier to misclassify them as machine-translated.

Machine-translated sentences, classified as Human-translated

The incorrect classification of machine-translated sentences as human-translated was positively impacted by the following features (Table 3.5):

- UPOS: `INTJ` (Interjection)
- Dependency relation: `flat:name` (names), `discourse` (interjections and other discourse particles and elements)
- NER: `PER` (person)
- Animacy: `Animate`

The presence of interjections and discourse markers may have signaled to the classifier that the sentence was more natural and conversational, leading to a misclassification. Similarly, `flat:name`, which is often used for multi-word names or set expressions in names, could have been misinterpreted as a hallmark of human translation. The classifier may have learned that proper handling of names — particularly those with correct transliteration — typically indicates human involvement. However, modern MT systems are capable of producing exclamations or interjection-like tokens when the source text includes them ("Wow!", "Oh," etc.). They also can replicate or preserve named entities quite smoothly, accidentally fooling the classifier.

Additionally, the presence of animate morphological marking (e.g., animate accusative endings for masculine nouns or pronoun usage that signals animacy) pushed the classifier toward labeling the text as human-translated. The classifier may have interpreted the correct handling of animacy as a more advanced morphological choice, typically expected from human translators. However, contemporary MT systems are increasingly capable of handling animacy correctly, which may have led to misclassification.

Conversely, the classifier was less likely to assign the human-translated label when the following features were present:

- UPOS: `CCONJ` (coordinating conjunction), `DET` (determiner), `VERB`
- Dependency relation: `acl:relcl` (relative clause modifier), `det` (determiner), `fixed` (fixed multiword expression)
- NER: `NONE`, `ORG` (organization)

In previous sections, `CCONJ` (Coordinating Conjunctions) and `DET` (Determiners) were identified as features commonly associated with human translations. However, in the current analysis, their presence pushed the classifier toward the machine-translated label. This may be explained by a lack of variation in their use within the misclassified sentences. When conjunctions and determiners appear in a repetitive manner, the classifier may have perceived them as indicative of machine translation.

Additionally, fixed multi-word expressions were previously identified as features that positively impacted machine translation classification. Their presence in this category is therefore consistent with earlier findings — suggesting that the classifier associates formulaic, standardized expressions with machine-translated output.

Key takeaways

Human-translated sentences labeled as "machine-translated" often show:

- Frequent adjectives or fixed phrases, which the classifier incorrectly equates with a literal or formulaic machine translation style.
- A lower occurrence of adverbials and subordinate clauses, making the text appear syntactically simpler—causing the classifier to misinterpret it as machine-generated.

Machine-translated sentences labeled as "human-translated" tend to display:

- Interjections, discourse markers, or well-formed personal names (NER = `PER`), which can look convincingly natural.
- Fewer advanced syntactic and morphological features, such as varied verb usage, relative clauses, and determiners. The classifier may have mistaken the absence of features typically associated with machine translation for a sign of human involvement.

Conclusion

In this thesis, my goal was to investigate whether and how modern machine translation output can be systematically distinguished from human translations in two specific language pairs: English–Russian and English–Czech. Despite recent advances in neural MT systems, subtle linguistic indicators can still reveal the "machine-like" characteristics embedded in machine-translated text. Leveraging a multilingual language model (XLM-R) fine-tuned for a binary classification task, I explored data from the WMT conference (2020–2022) and performed extensive experiments to assess how effectively it could classify human versus machine translations. In addition, I employed Integrated Gradients interpretability method to identify which linguistic features most strongly influence the classifier's decisions.

Those models from the 2020–2022 period were specialized encoder-decoder architectures, unlike current state-of-the-art approaches that utilize large language models, often fine-tuned specifically for translation-related tasks Kocmi et al. (2024). Therefore, the findings presented in this thesis may not generalize to the new generation of MT systems.

A key component of the research design was the creation of labeled datasets from both Direct Assessment and COMET-based quality metrics. Each method produced a distribution of translation "quality scores", which I further stratified to determine whether classification would be easier or harder, when MT quality neared human-like fluency. I initially used a simple baseline classifier (logistic regression with TF-IDF vectors) as a point of comparison. Although it performed modestly across various score ranges, the baseline underscored the inherent difficulty of discriminating between high-quality MT output and human translations.

The experiments revealed that fine-tuning XLM-R on selectively filtered datasets (particularly those excluding near-perfect MT scores) produced the most robust classification performance. However, I observed that the classifier struggled more as MT scores approached human-like quality. Modern translation systems reached an impressive level of quality, and distinguishing high-scored machine output from human work proved especially difficult for my classifier. The easiest classification was achieved using a COMET-based dataset in the medium-quality range (75–85), suggesting that mixed-quality machine outputs provide stronger discriminative features. Here, it achieved a ROC-AUC of around 0.70 on the test set, representing a substantial improvement over the baseline approach.

Beyond raw classification metrics, I used Integrated Gradients method and subsequent statistical analyses to investigate how the classifier determined whether a given sentence was likely produced by a human or a machine. By mapping subword-level attributions to broader linguistic features — such as part-of-speech (POS) tags, grammatical case, morphological agreement, and named entities — I discovered the following key outcomes:

- **Morphological Complexity**: In morphologically rich languages like Czech and Russian, correct and contextually nuanced handling of cases, genders, and animacy often signaled human translation.
- **Syntactic Variation**: Human-translated sentences tended to display more

flexible use of relative clauses, adverbial phrases, and discourse particles, whereas machine translations often featured rigid adjective–noun patterns.

- **Named Entity Adaptation**: Proper handling of organization or location names — particularly with respect to transliteration conventions — strongly indicated human translation.

- **Lexical and Stylistic Markers**: Overreliance on certain pronouns, conjunctions, or fixed expressions was often a hallmark of more "machine-like" direct translations, whereas nuanced use of intensifiers and domain-specific terms was characteristic of human style.

Additionally, the analysis identified differences between Czech and Russian translations. In Czech translations, relative clause modifiers and temporal expressions related to months and days played a significant role in indicating human translations, while simpler adjective-noun constructions were typical for machine translations. Russian translations, on the other hand, placed greater emphasis on the handling of named entities, such as organizations and locations. Machine translations frequently demonstrated direct adjective-noun pairings and kept foreign words unchanged, whereas human translations were distinguished by their accurate adaptation of location names and more varied noun-phrase structures.

Although the classifier showed an encouraging ability to detect MT output, especially in medium-quality ranges, it struggled in edge cases — namely, extremely high-quality MT that closely mimics human fluency and sometimes even lower-quality human translations that adopt literal choices. This difficulty highlights how near-human MT output is increasingly blurring boundaries. The WMT data also comes mostly from news-oriented texts, limiting exposure to other domains such as legal or literary translations. Nevertheless, my findings confirm that a fine-tuned multilingual language model, coupled with carefully constructed datasets and interpretability techniques, outperforms simpler baselines in spotting subtle indications of machine translation.

Suggestions for Further Work

1. **Expanding Language Pairs and Domains**
   While I focused on English–Russian and English–Czech, an obvious next step is to extend this approach to other language pairs and more varied text domains. Low-resource languages might present new challenges for both MT systems and human translators, potentially revealing different features and error patterns.

2. **Incorporating Context-Aware Features**
   My classification approach considered individual sentences in isolation. However, MT errors (and human stylistic choices) often become more apparent in multi-sentence contexts — such as paragraph-level coherence or consistency in tense usage. A classifier that aggregates evidence across larger segments could potentially catch discourse-level anomalies and further boost accuracy.

3. **Domain Adaptation and Fine-Tuning**
   Since translation style can shift dramatically across genres and domains,

future research might investigate whether domain-adapted language models can spot domain-specific MT artifacts. An in-depth comparison of how domain adaptation affects both the classifier's performance and the learned linguistic signals would be an insightful extension of this work.

Bibliography

J. Bergstra, D. Yamins, and D. D. Cox. Making a science of model search: Hyperparameter optimization in hundreds of dimensions for vision architectures. *Proceedings of the 30 th International Conference on Machine Learning*, 28: 115–123, 2013.

A. Binder, G. Montavon, S. Bach, K.-R. Müller, , and W. Samek. Layer-wise relevance propagation for neural networks with local renormalization layers. *arXiv:1604.00825*, 2016.

A. Conneau, G. Lample, R. Rinott, H. Schwenk, V. Stoyanov, A. Williams, and S. R. Bowman. Xnli: Evaluating cross-lingual sentence representations. *arXiv:1809.05053v1*, 2018.

A. Conneau, K. Khandelwal, K. Lee, V. Chaudhary, G. Wenzek, F. Guzmán, E. Grave, M. Ott, L. Zettlemoyer, and V. Stoyanov. Unsupervised cross-lingual representation learning at scale. *arXiv:1911.02116v2*, 2020.

J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. *arXiv:1810.04805v2*, 2019.

Y. Graham, T. Baldwin, and N. Mathur. Accurate evaluation of segment-level machine translation metrics. *Proceedings of the 2015 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 1183–1191, 2015.

D. Ippolito, D. Duckworth, C. Callison-Burch, and D. Eck. Automatic detection of generated text is easiest when humans are fooled. *arXiv:1911.00650v2*, 2020.

K. S. Jones. A statistical interpretation of term specificity and its application in retrieval. *Journal of Documentation*, 28:11–21, 1972.

T. Kocmi, E. Avramidis, R. Bawden, O. Bojar, A. Dvorkovich, C. Federmann, M. Fishel, M. Freitag, T. Gowda, R. Grundkiewicz, B. Haddow, M. Karpinska, P. Koehn, B. Marie, C. Monz, K. Murray, M. Nagata, M. Popel, M. Popović, M. Shmatova, S. Steingrímsson, and V. Zouhar. Findings of the wmt24 general machine translation shared task: The llm era is here but mt is not solved yet. *Proceedings of the Ninth Conference on Machine Translation*, pages 1–46, 2024.

G. Lample and A. Conneau. Cross-lingual language model pretraining. *arXiv:1901.07291v1*, 2019.

K. P. Murphy. *Probabilistic Machine Learning*. Adaptive Computation and Machine Learning. The MIT Press, Cambridge, Massachusetts, 2022. ISBN 978-0262-04682-4.

H. H. Nguyen, P. N. Adams, and S. J. Miller. *Building Statistical Models in Python*. Packt Publishing Ltd., Birmingham, UK, 2023. ISBN 978-1-80461-428-0.

K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu. Bleu: a method for automatic evaluation of machine translation. *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 311–318, 2002.

R. Rei, C. Stewart, A. C. Farinha, and A. Lavie. Comet: A neural framework for mt evaluation. *arXiv:2009.09025v2*, 2020.

R. Rei, J. G. C. de Souza, D. M. Alves, C. Zerva, A. C. Farinha, T. Glushkova, A. Lavie, L. Coheur, and A. F. T. Martins. Comet-22:vunbabel-ist 2022 submission for the metrics shared task. *Proceedings of the Seventh Conference on Machine Translation (WMT)*, pages 578–585, 2022.

A. Shrikumar, P. Greenside, and A. Kundaje. Learning important features through propagating activation differences. *arXiv:1704.02685*, 2017.

M. Straka. Udpipe 2.0 prototype at conll 2018 ud shared task. *Proceedings of the CoNLL 2018 Shared Task: Multilingual Parsing from Raw Text to Universal Dependencies*, pages 197–207, 2018.

J. Straková, M. Straka, and J. Hajic. Neural architectures for nested NER through linearization. *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, 28:5326–5331, 2019.

M. Sundararajan, A. Taly, and Q. Yan. Axiomatic attribution for deep networks. *arXiv:1703.01365v2*, 2017.

R. F. Tate. Correlation between a discrete and a continuous variable. point-biserial correlation. *The Annals of Mathematical Statistics*, 25:603–607, 1954.

S. Vasishth, D. Schad, A. Bürki, and R. Kliegl. *Linear Mixed Models in Linguistics and Psychology: A Comprehensive Introduction.* Potsdam, Germany, 2022.

A. Wang, A. Singh, J. Michael, F. Hill, O. Levy, and S. R. Bowman. Glue: A multi-task benchmark and analysis platform for natural language understanding. *arXiv:1804.07461v3*, 2019.

List of Figures

List of Tables

List of Abbreviations

MT – Machine Translation
XLM-R – XLM-RoBERTa
DA – Direct Assessment
GLUE – General Language Understanding Evaluation
XNLI – Cross-lingual Natural Language Inference
MLM - Masked Language Model
ROC – Receiver Operating Characteristic
AUC – Area Under Curve
TPR – True Positive Rate
FPR – False Positive Rate
LRP – Layer-wise Relevance Propagation
IG – Integrated Gradients
WMT – Workshop on Machine Translation
TF-IDF – Term Frequency-Inverse Document Frequency
TPE – Tree-structured Parzen Estimator
OLS – Ordinary Least Squares
POS – Part of Speech
UD – Universal Dependencies
NE – Named Entity
NER – Named Entity Recognition
NLP – Natural Language Processing
UPOS – Universal Part of Speech

A Attachments

A.1 Experiments visualization

A.1.1 Metrics visualization for DA datasets experiments

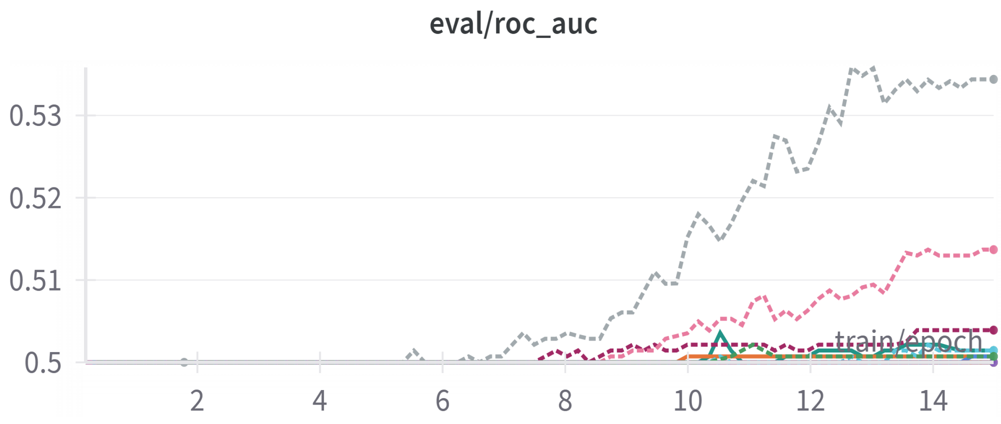

Figure A.1 ROC-AUC score for (60–70) DA dataset.

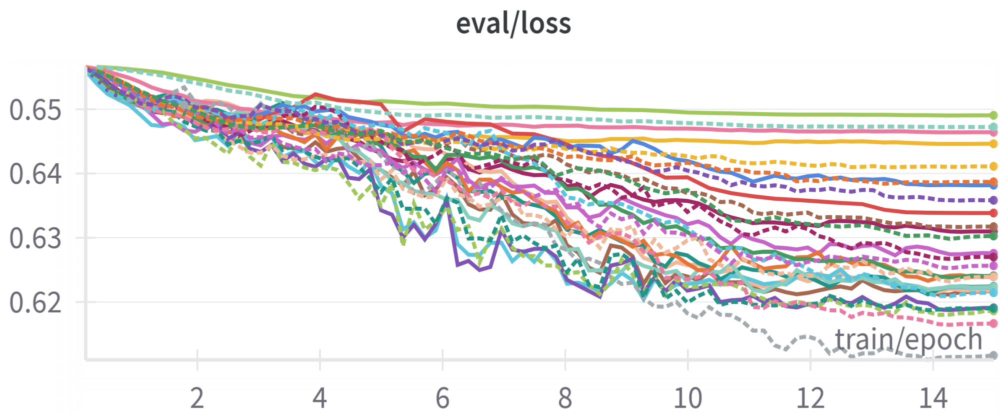

Figure A.2 Evaluation loss for (60–70) DA dataset.

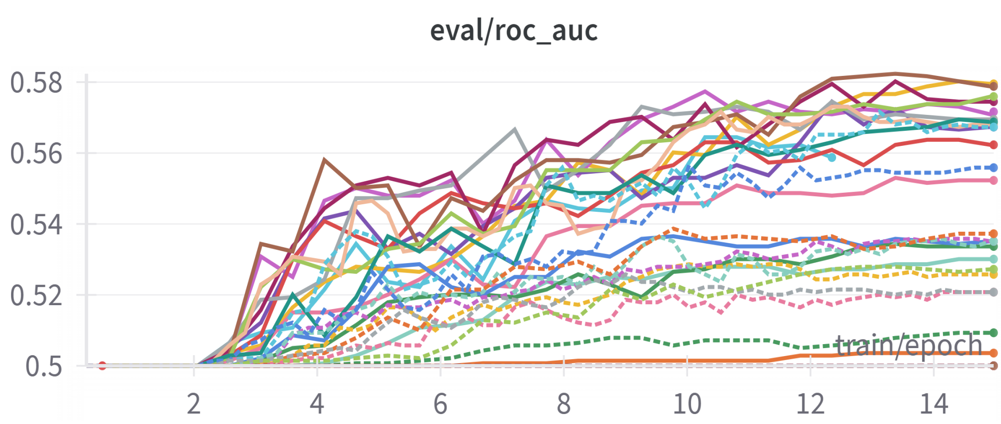

Figure A.3 ROC-AUC score for (70–80) DA dataset.

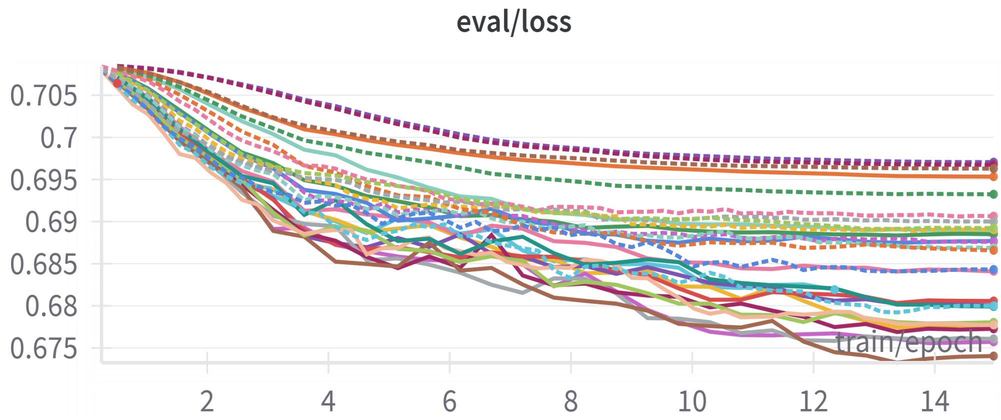

Figure A.4 Evaluation loss for (70–80) DA dataset.

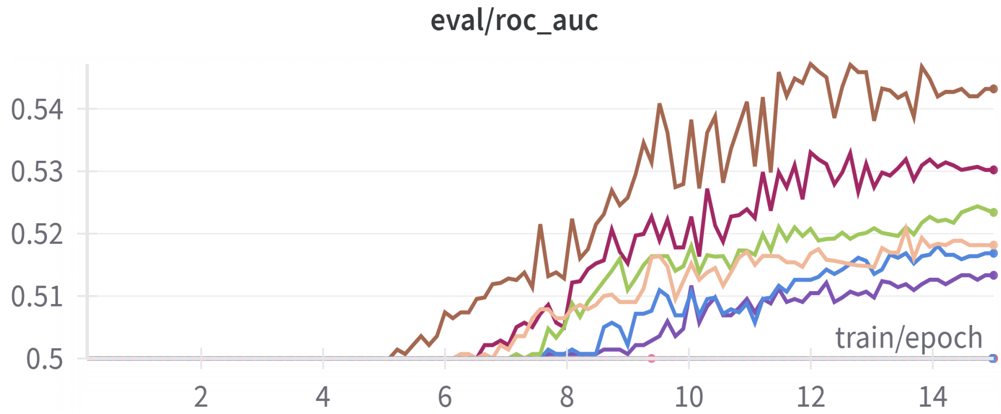

Figure A.5 ROC-AUC score for (80–90) DA dataset.

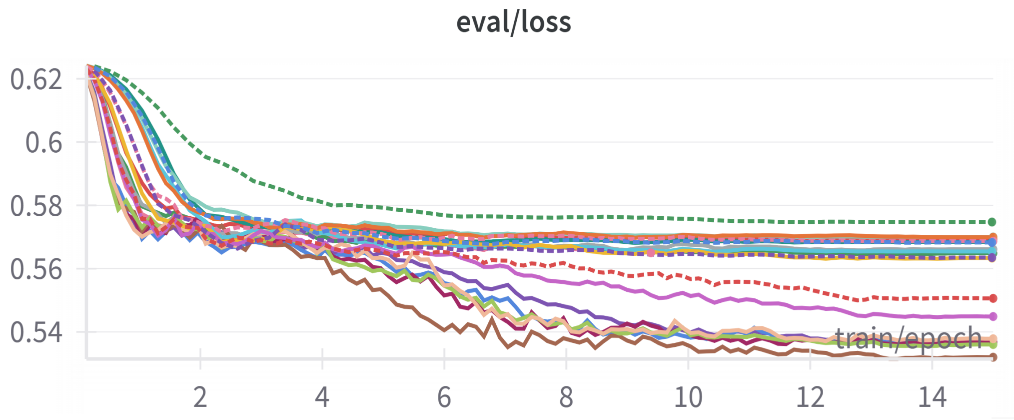

Figure A.6 Evaluation loss for (80–90) DA dataset.

A.1.2 Metrics visualization for COMET datasets experiments

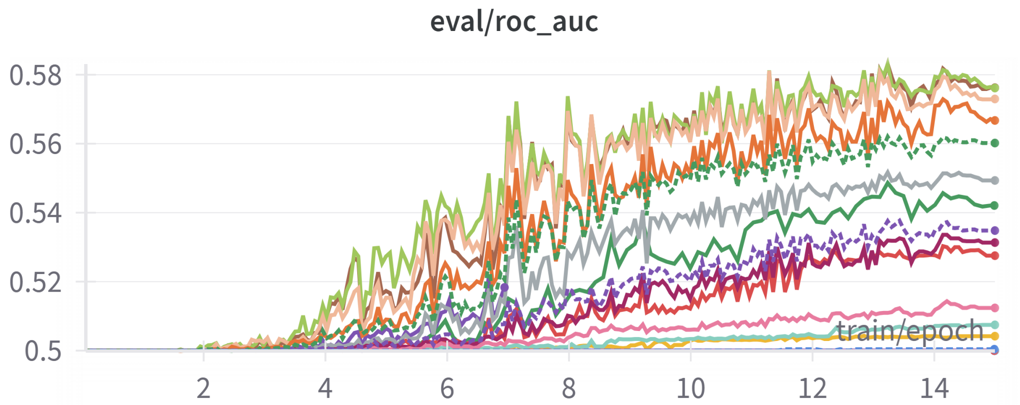

Figure A.7 ROC-AUC score for (85–95) COMET dataset.

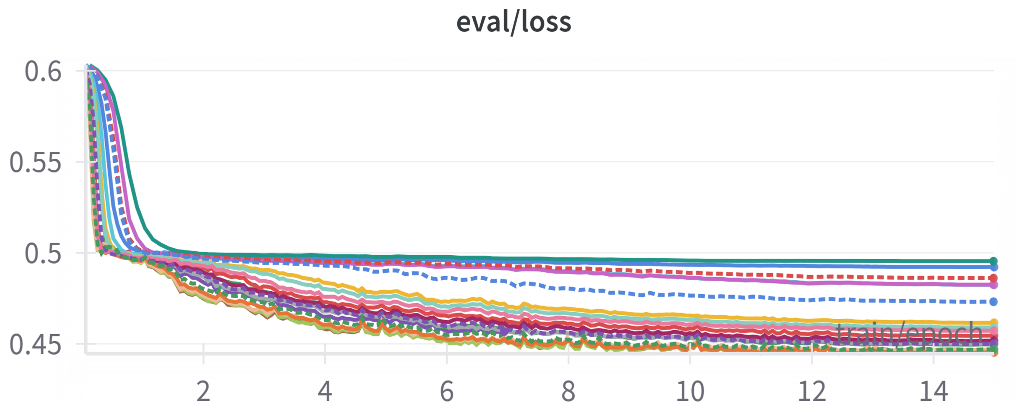

Figure A.8 Evaluation loss for (85–95) COMET dataset.

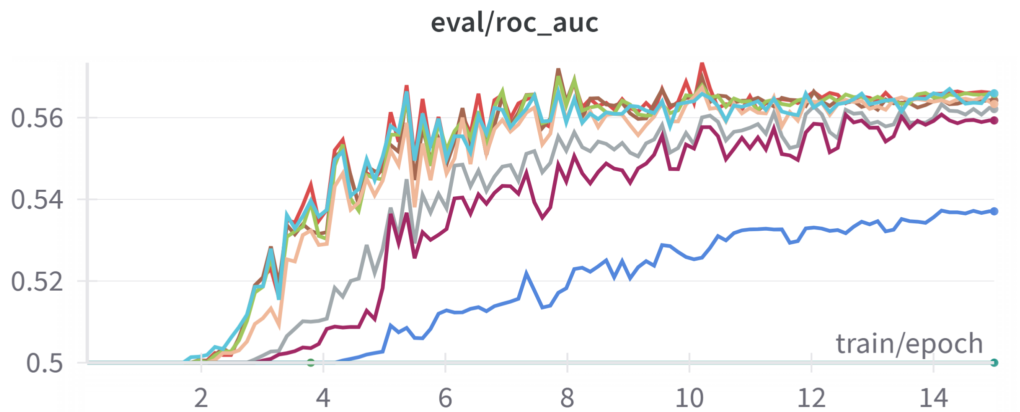

Figure A.9 ROC-AUC score for (95–100) COMET dataset.

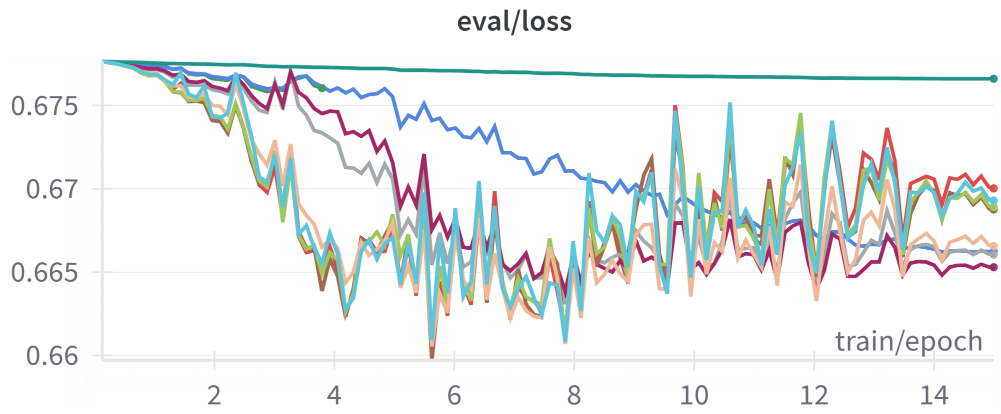

Figure A.10 Evaluation loss for (95–100) COMET dataset.

A.2 Transformer-interpret package custom changes

This issue arises from the handling of the `token_type_ids` tensor, which is used to indicate token types corresponding to the `input_ids`.

In transformer models like BERT, the `token_type_ids` tensor serves to distinguish between tokens originating from different input segments. For instance, when a BERT model processes two sentences, `token_type_ids` assigns a value of 0 to tokens from the first sentence and a value of 1 to tokens from the second. This mechanism allows the model to differentiate the contexts of each segment. However, the XLM-R model is architecturally different. Its embedding layer for token types is defined as:

(`token_type_embeddings`): Embedding(1, 1024)

Here, the embedding dictionary size is 1, meaning the model does not utilize `token_type_ids` in the same way as BERT.

In its original implementation, the transformer-interpret package initializes the `token_type_ids` tensor with zeros and ones, which is incompatible with XLM-R's embedding layer. To resolve this issue, I modified the tensor initialization, ensuring that it is a zero-only tensor. This adjustment aligns with the XLM-R model's requirement and prevents the aforementioned error. The updated code for the relevant function is included in electronic attachment A.4.

A.3 List of linguistic tags

UPOS

NUM – Numerical expressions
CCONJ – Coordinating conjunction
NOUN – Noun
PRON – Pronoun
ADJ – Adjective
DET – Determiner
PART – Particle
ADP – Adposition
INTJ – Interjection

VERB – Verb

Dependency relations

acl:relcl – Relative clause modifier
nummod – Numeric modifiers
advmod:emph – Emphasizing word, intensifier
cc – Coordinating conjunction
ccomp – Clausal complement
obj – Object
obl – Oblique modifier
amod – Adjectival modifier
appos – Appositional modifier
flat:foreign – Foreign words
conj – Conjunct
det – Determiner
fixed – Fixed multi-word expression
mark – Subordinating marker
flat:name – Names
nmod – Nominal modifier
advcl – Adverbial clause modifier
advmod – Adverbial modifier
case – Case marking
flat – Flat expression
obl:arg – Oblique Argument
cop – Copula
discourse – Interjections and other discourse particles and elements

NER

T|tm – Time:Months
T|td – Time:Days
ORG – Organization
LOC – Location
PER – Person
P|ps – Personal Name:Surname
i_ – Institution
gc – States

Case

Acc – Accusative

A.4 Electronic attachments

The electronic attachment includes code for model fine-tuning and analysis, datasets and annotated test data. It has the following structure:

/src

`analysis.ipynb` - this Jupyter notebook was used for correlation analysis using OLS regression and point-biserial correlation coefficient.

`baseline.ipynb` - this Jupyter notebook was used for training and evaluation of baseline model (logistic regression).

`explainer-function-fix.py` - a function from transformer-interpret package, adapted to work with the XLM-R model.

`requirements.txt` - list of required packages for running `analysis.ipynb`

`run.sh` - this file was used for running `xlm-r-fine-tuning.py` on GPU cluster.

`xlm-r-fine-tuning.py` - this code was used for XLM-R_{base} model fine-tuning.

/data - contains test data annotated with linguistic features and attribution score, used for analysis.

/annotated-data

`annotation-correct-hm-cs.csv`
`annotation-correct-hm-ru.csv`
`annotation-correct-mt-cs.csv`
`annotation-correct-mt-ru.csv`
`annotation-incorrect-hm-cs.csv`
`annotation-incorrect-hm-ru.csv`
`annotation-incorrect-mt-cs.csv`
`annotation-incorrect-mt-ru.csv`

/datasets - contains datasets created based on DA and COMET metric, grouped based on quality and splitted to training, test, and validation subsets.

/comet-scored-data

/75-85

`test-75-85.csv`
`training-75-85.csv`
`validation-75-85.csv`

/85-95

`test-85-95.csv`
`training-85-95.csv`
`validation-85-95.csv`

/95-100

`test-95-100.csv`
`training-95-100.csv`
`validation-95-100.csv`

/da-scored-data

- **/60-70**
  - `test-60-70.csv`
  - `training-60-70.csv`
  - `validation-60-70.csv`
- **/70-80**
  - `test-70-80.csv`
  - `training-70-80.csv`
  - `validation-70-80.csv`
- **/80-90**
  - `test-80-90.csv`
  - `training-80-90.csv`
  - `validation-80-90.csv`
- **/90-100**
  - `test-90-100.csv`
  - `training-90-100.csv`
  - `validation-90-100.csv`